

Ganit Yadav

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Education

VIT Bhopal University, Madhya Pradesh

B.Tech in Electronics and Communication Engineering

Sept 2023 – Ongoing

CGPA: 9.3/10

Govt. Boys H.S. School, Sidhi (M.P.)

Apr 2023

Class XII

Percentage: **94.2%**

Govt. Boys H.S. School, Sidhi (M.P.)

Apr 2021

Class X

Percentage: **80%**

Technical Skills

- **Languages:** C++, Python, Embedded C.
- **Coursework:** Embedded Systems, Internet of Things (IoT), Microcontroller & Microprocessor, Digital Electronics.
- **Tools & Software:** MATLAB, LTspice, Keil uVision5, Arduino IDE, VSCode.

Experience

Software Engineering Intern — TVS Motor Company

May 2026 – July 2026

Developed **SpectraX**, a Python-based CAN Trace Analysis Tool for automotive trace data analysis. Implemented DBC-based signal decoding, timing analysis, multi-trace comparison, trace file conversion, interactive visualization, and optimized caching techniques for efficient processing of large CAN trace files.

[Certificate](#)

Embedded System Design Internship — Maven Silicon

May 2025 – July 2025

Developed the “Obstacle Detection for Visually Impaired” project using Arduino and ultrasonic sensors with buzzer alerts. Gained hands-on experience in embedded programming, sensor integration, and user-centric IoT design.

[Certificate](#)

Academic Projects

Visitor Notification and Smart Locking System | *ESP32-CAM, Telegram Bot, LCD, Solenoid Lock* **Jan 2025 - May 2025**

- Developed an IoT-based visitor notification system using ESP32-CAM to capture visitor images and transmit them securely to a mobile device via a Telegram Bot.
- Implemented a push-button-based image capture mechanism, while enabling remote door locking and unlocking exclusively through Telegram commands (/lock, /unlock).
- Integrated an external LCD display to show real-time availability status and custom messages, enhancing visitor communication and user awareness. [GitHub](#)

Smart Irrigation Monitoring System | *ESP32, IoT, Blynk, DHT11, Soil Moisture Sensor*

Sep 2024 - Dec 2024

- Designed and developed an IoT-based smart irrigation system using ESP32 to monitor soil moisture and environmental parameters in real time.
- Implemented automatic motor ON/OFF control based on predefined threshold values of soil moisture and DHT11 sensor readings to ensure optimal irrigation.
- Enabled remote monitoring and control through the Blynk IoT platform, allowing users to track sensor data and system status in real time. [GitHub](#)

Obstacle Detection System for Visually Impaired | *Arduino, Ultrasonic Sensor, Buzzer*

May 2025 – July 2025

- Developed an embedded system-based assistive solution for visually impaired individuals to detect nearby obstacles using ultrasonic sensing technology.
- Implemented real-time distance measurement by calculating echo time of ultrasonic waves reflected from obstacles.
- Designed a distance-based alert mechanism where buzzer frequency dynamically increases as the obstacle distance decreases, providing intuitive real-time feedback. [GitHub](#)

Certifications

Introduction to Internet of Things – NPTEL

[Certificate](#)

C Programming – NPTEL

[Certificate](#)